NatureLM-audio User Manual

Cane Toad Vocalisation Detection Using NatureLM-audio

User manual prepared to support the implementation and execution of

NatureLM-audio for cane toad vocalisation detection.

Group 19

Axl Espiritu	S387016
A N M Shahariyar	S391094
Jay Malcampo	S387478
Simin Li	S390115

Supervisor:

Dr. Yakub Sebastian

CHARLES DARWIN UNIVERSITY

FACULTY OF SCIENCE AND TECHNOLOGY

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Introduction

This manual provides instructions for installing, configuring and running NatureLM-audio for cane toad vocalisation detection. The procedures described in this guide reflect the workflow used during the project and include both local Windows execution and Google Colab deployment.

For the latest installation requirements and software updates, refer to the official NatureLM-audio repository:

https://github.com/earthspecies/NatureLM-audio

Requirements

Hardware

Minimum Requirements

- Windows 10 or Windows 11
- Multi-core CPU
- Internet connection

Recommended Requirements

- NVIDIA GPU with more than 16 GB VRAM
- 32 GB RAM
- SSD storage

Software

- Python 3.10+
- Git
- Windows PowerShell
- Google Account
- Hugging Face Account

Model Access Requirements

NatureLM-audio requires access to Meta Llama 3.1 8B Instruct through Hugging Face.

Before proceeding:

1. Create a Hugging Face account.

2. Request access to Meta Llama 3.1 8B Instruct: https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct
3. Wait for approval.
4. Generate a Hugging Face access token.
5. Store the token securely.

Local Environment

Local CPU Inference (Powershell)

Clone the Repository

Open Windows PowerShell and run:

git clone https://github.com/earthspecies/NatureLM-audio.git

cd NatureLM-audio

Create a Virtual Environment

python -m venv venv

venv\Scripts\activate

Install Dependencies

Follow the installation instructions provided in the official NatureLM-audio repository.

Project-Specific Dependency Notes

During implementation, several dependency conflicts were encountered. The following package versions were successfully used during the project and may be useful if installation issues occur.

pip install transformers==4.44.2

pip install tokenizers==0.19.1

pip install peft==0.11.1

pip install accelerate==0.33.0

pip install librosa

pip install soundfile

pip install sentencepiece

pip install mir_eval

pip install resampy

If package conflicts occur, install the versions above and restart the environment before continuing.

Run Command-Line Inference

Example:

```
uv run naturelm infer \
--cfg-path configs/inference.yml \
--audio-path assets \
--query "Caption the audio" \
--window-length-seconds 10.0 \
--hop-length-seconds 10.0
```

Output:

inference_output.jsonl

Local Web Inference

Run Web Inference

Launch the browser-based interface:

```
uv run naturelm inference-app \
--cfg-path configs/inference.yml \
--merging-alpha 0.5
```

After launching, NatureLM-audio generates:

- Local access link
- Public access link

Example:

Running on local URL:
http://127.0.0.1:7860

Running on public URL:
https://xxxxxxxx.gradio.live

Open either link using a web browser.

To perform inference:

1. Upload an audio file.
2. Enter a prompt.
3. Submit the request.
4. Review the generated response.

Supported formats:

- WAV
- FLAC

- MP3
- OGG

Local GPU Testing (PowerShell)

This section verifies whether the local GPU can run NatureLM-audio. Successful execution depends on the available GPU memory and CUDA configuration.

Check whether an NVIDIA GPU is available:

nvidia-smi

Verify CUDA support:

import torch

print(torch.cuda.is_available())

If the GPU contains sufficient VRAM and NatureLM-audio loads successfully, users may proceed with inference using either:

- Command-line inference
- Web inference

Example:

uv run naturelm infer \

--cfg-path configs/inference.yml \

--audio-path assets \

--query "Caption the audio" \

--window-length-seconds 10.0 \

--hop-length-seconds 10.0

or

uv run naturelm inference-app \

--cfg-path configs/inference.yml \

--merging-alpha 0.5

If CUDA is unavailable or insufficient GPU memory is detected, use Google Colab instead.

Google Colab Environment

Create a Google Colab Notebook

Create a new notebook and enable GPU acceleration:

Runtime → Change Runtime Type → GPU

Preferred GPU:

- NVIDIA A100

Mount Google Drive

```
from google.colab import drive
drive.mount('/content/drive')
```

Create Project Folders

```
import os
BASE = "/content/drive/MyDrive/NatureLM_project"


folders = [
    "code",
    "audio_files",
    "outputs",
    "hf_cache"
]


for folder in folders:
    os.makedirs(os.path.join(BASE, folder), exist_ok=True)
```

Folder structure:

```
NatureLM_project/
│
├── code/
├── audio_files/
├── outputs/
└── hf_cache/
```

Clone the Repository

```
%cd /content/drive/MyDrive/NatureLM_project/code
!git clone https://github.com/earthspecies/NatureLM-audio.git
```

Configure Hugging Face

Set the cache location:

```
import os

os.environ["HF_HOME"] = \
"/content/drive/MyDrive/NatureLM_project/hf_cache"
```

Login:

```
from huggingface_hub import login

login()
```

Paste the Hugging Face access token when prompted.

Upload Audio Files

Store audio recordings in:

NatureLM_project/audio_files/

Supported formats:

- WAV
- FLAC
- MP3
- OGG

Enable GPU Runtime

Navigate to:

Runtime → Change Runtime Type → GPU

Preferred:

- NVIDIA A100

Run Inference

Execute the required inference notebook or batch-processing script.

Recommended workflow:

1. Upload audio recordings.

2. Configure prompts.
3. Execute inference.
4. Save outputs directly to Google Drive.

Batch-Processing

For large datasets:

1. Divide recordings into manageable batches.
2. Process one batch at a time.
3. Save outputs after each batch.
4. Continue with the next batch.

This reduces the risk of losing progress if a Colab session terminates unexpectedly.

Output Files

NatureLM-audio outputs are generated in JSONL format.

Example:

batch_001.jsonl

Each entry typically contains:

- Audio file path
- User prompt
- Generated response
- Predicted species information

Recommended output location:

NatureLM_project/outputs/

Generated JSONL files are automatically saved to this directory and can be downloaded directly from Google Drive.

Troubleshooting

HuggingFace Authentication Error

Cause:

- Access token missing or invalid.

Solution:

- Verify token permissions.
- Confirm access approval for Meta Llama 3.1.

Dependency Conflict Error

Cause:

- Package version incompatibilities.

Solution:

- Install the dependency versions listed in Section 3.3.
- Restart the environment.

Missing Python Packages

Cause:

- Dependencies not installed.

Solution:

- Reinstall missing packages.

CUDA Not Detected

Cause:

- CUDA installation issue.
- Unsupported GPU.

Solution:

- Verify GPU drivers.
- Confirm CUDA installation.
- Use Google Colab if necessary.

Out-of-Memory Error

Cause:

- Insufficient GPU memory.

Solution:

- Reduce batch size.

- Use higher-memory GPUs such as A100.

Colab Session Reset

Cause:

- Runtime timeout or disconnect.

Solution:

- Store all files in Google Drive.
- Save outputs after each completed batch.